

Lista 4 - Derivadas de funções x^n e $\sqrt[n]{x}$, Derivada de e^x e $\ln|x|$ e Derivadas de funções trigonométricas

Calcule

$$1. \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}, n \in \mathbb{R}_*^+$$

$$2. \lim_{h \rightarrow 0} \frac{\sqrt[n]{x+h} - \sqrt[n]{x}}{h}, n \in \mathbb{R}_*^+$$

$$3. \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}$$

$$4. \lim_{h \rightarrow 0} \frac{\ln|x+h| - \ln|x|}{h}$$

$$5. \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$6. \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$7. \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h}$$

$$8. \lim_{h \rightarrow 0} \frac{\sec(x+h) - \sec x}{h}$$

$$9. \lim_{h \rightarrow 0} \frac{\csc(x+h) - \csc x}{h}$$

$$10. \lim_{h \rightarrow 0} \frac{\cot(x+h) - \cot x}{h}$$

Nos exercícios de 11 a 30, calcule $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ sendo:

$$11. f(x) = x^4 + 3x^3 - \sqrt[3]{x}$$

$$12. f(x) = \frac{x^7}{7} + 12x^3 - x^{1/2}$$

$$13. f(x) = x^5 + \frac{7x^4}{16} - x^4$$

$$14. f(x) = \frac{x^9 - x^7}{18} + \frac{\sqrt[3]{x}}{8}$$

$$15. f(x) = x^{17} + 11\sqrt[11]{x} - 23\sqrt[7]{x}$$

$$16. f(x) = 3x^2 + 12x^{\frac{1}{2}} - \frac{3x^9}{2}$$

$$17. f(x) = \frac{28x^4}{3} + 5x^{\frac{2}{3}} - \frac{22x^3}{5}$$

$$18. f(x) = \sqrt[4]{16x} + 8x^{1/7} + \frac{8x^2}{5}$$

$$19. f(x) = \sqrt[2]{35x} + \frac{x^2}{8} - \frac{7x^9}{11}$$

$$20. f(x) = 35x^{\frac{1}{7}} + \frac{24x^2}{18} - 13\sqrt[45]{x}$$

$$21. f(x) = e^x - 4x^{16} - \frac{4e^x}{12}$$

$$22. f(x) = \frac{e^x}{5} + \sqrt[4]{12x} - \frac{e^x}{9}$$

$$23. f(x) = \frac{7e^x + \ln|x| - 3\ln|x|}{3}$$

$$24. f(x) = \frac{e^x - x^3}{3} + \frac{\ln|x|}{5}$$

$$25. f(x) = \frac{x^{21}}{e} + e^{11} - \frac{4\ln|x|}{3}$$

$$26. f(x) = \frac{\sin x - \cos x}{2}$$

$$27. f(x) = \frac{3x^4}{8} + 5\cos x - \frac{\sqrt[4]{x}}{7}$$

$$28. f(x) = 13\sqrt[2]{x} + 8\tan x + \frac{10\sin x}{7}$$

$$29. f(x) = \sqrt[3]{23x} + \frac{3\sin x + 2\sec x}{13}$$

$$30. f(x) = 7\sqrt[13]{x} + \frac{\cot x + 13\ln|x|}{3} - \frac{e^x + 2\ln|x|}{5}$$

Nos exercícios de 31 a 40, encontre a equação da reta tangente a função $f(x)$ no ponto p , sendo:

31. $f(x) = \sqrt[3]{x} - \frac{x^4 - e^x}{4}, p = 4$

32. $f(x) = \frac{3\ln|x|}{5} + 2x^5 + \sin x, p = \frac{\pi}{2}$

33. $f(x) = 4x^{\frac{1}{4}} + \tan x + \frac{5e^x}{2}, p = \frac{\pi}{2}$

34. $f(x) = 5 \cot x + \frac{11\ln|x|}{5} - \frac{5e^x + \cos x}{5}, p = \frac{\pi}{2}$

35. $f(x) = \sin x - \cos x + \frac{3e^x - \ln|x|}{\pi}, p = \pi$

36. $f(x) = 3\sqrt[3]{x} - \frac{2x^{\frac{3}{2}}}{3} + \frac{\tan x}{5}$

37. $f(x) = \frac{3e^x - \ln|x| + x^{\frac{3}{5}}}{e}, p = 1$

38. $f(x) = \sqrt[3]{x} + 4\sqrt[5]{x} - \frac{5x^{\frac{2}{5}}}{2}, p = 4$

39. $f(x) = \sin x - 3 \cos x + 4 \tan x - e^x, p = \frac{\pi}{2}$

40. $f(x) = \frac{x^{\frac{3}{7}} + 3\ln|x| - e^x + 3x^{\frac{1}{3}}}{e^3}$